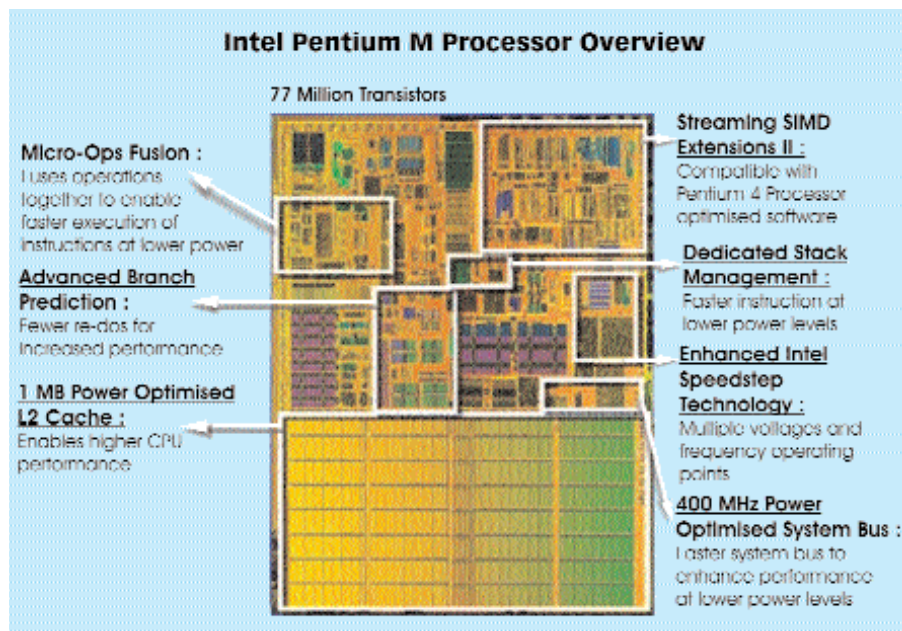


the overall latency in the system, thus contributing to higher performance with low power consumption. The second—Dedicated Stack Manager—reduces the overall number of micro-operations required to generate higher performance at a lower power.

The new chip does not compromise on performance—it completes more instructions per clock cycle (IPC). Besides, the Pentium-M has a 1 MB Level 2 cache—twice that of the P4's 512 KB L2 cache. The Israel-based design team, headed by Mooly Eden, realised that each and every processor clock cycle costs valuable battery power. Hence the Banias was designed for a high IPC rate, so that it could do a lot of work in fewer clock cycles. Banias thus has a shorter pipeline, and runs at lower clock frequencies than the clock speed racer Pentium 4.

The processor is manufactured using Intel's industry-leading 0.13-micron process technology, and consists of 77 million transistors. It also includes a 400 MHz power optimised system bus, a 1 MB low-power L2 cache (which turns off parts of the high-speed memory when it's not needed), and technology to reduce overall platform power consumption. It also provides support for Enhanced Intel SpeedStep technology (that allows it to drop to multiple lower speed settings) with multiple voltage and frequency operating points.

The Pentium-M notebooks offer four to five power steps as opposed to the two speeds (fast and slow) of the original SpeedStep notebooks. It also supports



Streaming SIMD Extensions II. This makes the Pentium-M fully compatible with P4-optimised software, although there's no Hyper-Threading support.

Slimming the laptop

To fit high-performance processors into all PC designs, including slim and ultra-slim notebooks, Intel Centrino mobile technology uses the Micro FCPGA (Flip Chip Pin Grid Array) and Micro FCBGA (Flip Chip Ball Grid Array) packaging technology. This technology has been bettered for a range of thinner, lighter notebook PC designs that can be less than an inch thick.

Thermal management is equally

essential to enabling thinner, lighter form factors. Intel engineers have designed the new notebook to further minimise thermal power, the target power level at which mobile systems work best and use the least amount of energy.

Other technology features that enable computer makers to design sleeker, smaller notebooks, sub-notebooks and tablet PCs include the special Low Volt (LV) and Ultra Low Volt (ULV) options. These options set the processor to run at a lower voltage specification, which is critical to lowering thermal power in tablet designs and extra-slim notebooks that are less than one-inch thick. And that means you can have a truly

AMD Follows Suit

AMD sprang a surprise on Intel by introducing 12 new mobile processors for thin-and-light and high-performance notebooks at CeBIT on March 12. The chips, to be sold under the Athlon XP-M brand, include three based on the recently released Barton processor for performance desktops, and five energy-efficient models for so-called thin and light notebooks that use a maximum of 25 watts of energy, according to AMD officials. Energy efficiency will help PC makers add wireless networking.

The Athlon XP-M family breaks down into three segments: a high performance segment based around the Barton processor; a middle segment based around the

standard Athlon XP chip; and the low-power segment which relies on a standard



Athlon XP that uses less electricity. The Barton chips will be sold with model numbers of 2200+, 2400+ and 2500+. Barton chips differ from other Athlon XP and Athlon XP-M chips, primarily in that they doubled the L2 cache, an

on-chip reservoir of memory for rapid data access.

AMD Athlon XP-M processors are designed with an open architecture, helping to ensure that the 802.11 wireless solutions from leading companies can be easily integrated into AMD processor-based mobile systems. The energy-efficient chips will pro-

vide less performance (very unlike the Centrino that is a whole new architecture), but they will let notebooks run longer on a battery charge. AMD is also working on a chip that consumes even less power, relying on a maximum of 16 watts, according to a representative, but it isn't ready yet.

The mobile AMD Athlon XP-M processors are compatible with AMD's Socket A infrastructure, and support the advanced 266 MHz AMD Athlon front-side bus. The low-voltage mobile AMD Athlon XP-M processors are pin compatible with Socket A infrastructure, but are packaged in a smaller uPGA package. Mobile AMD Athlon XP-M processors are manufactured using AMD's 0.13 micron copper process technology.